

Komorebi-Mediated Ambient Notifications for Calm Technology

Mamoru Hiramatsu
Graduate School of Information Science
Meisei University
Hino-shi, Tokyo, Japan
24mj003@stu.meisei-u.ac.jp

Toshitaka Amaoka
Graduate School of Information Science
Meisei University
Hino-shi, Tokyo, Japan
amaoka@is.meisei-u.ac.jp

Abstract

In today's information-overloaded society, Calm Technology—a design concept that presents essential information as an ambient part of the environment without demanding focused attention—is increasingly important. This research is inspired by the unique characteristic of *komorebi* (dappled sunlight), a Japanese term for sunlight filtering through trees, to gently capture attention while seamlessly blending into a space and environment. We propose a notification method that uses *komorebi* as its medium. To demonstrate our proposed method, we developed prototypes that realize both active notifications, where the system prompts for attention, and passive notifications, which users check voluntarily. The active notifications are implemented through a mechanical approach, while the passive notifications are achieved with an optical approach. The mechanical approach controls the orientation of a branch-and-leaf unit to alter the position and movement of *komorebi*. The optical approach modifies shadow hardness by switching between light sources.

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1 Introduction

In modern society, individuals are constantly exposed to a vast amount of information. Notifications from digital devices like smartphones and personal computers incessantly fragment our attention, potentially impairing cognitive control and concentration [Upshaw et al. 2022]. In response to this issue, Calm Technology proposed by Weiser and Brown, has gained attention as a design concept that avoids overburdening human attention by presenting minimal necessary information as part of the environment [Weiser and Brown 1996]. Building on this, Amber Case argues that information presentation should exist in the "periphery of perception" rather than at its "center" [Case 2015]. This implies that the notification medium itself should blend into the space, conveying information through subtle changes. From this perspective, natural phenomena are promising media for Calm Technology. Therefore, we focus on *komorebi* (see Figure 1), a Japanese term referring to sunlight

filtering through leaves. This concept is particularly well suited to Calm Technology. Architecturally, *komorebi* casts soft, dappled light indoors that maintains a tranquil atmosphere; psychologically, it evokes a sense of unity with nature and the passage of time, as its subtle variations, perceived in peripheral vision, can be naturally elevated to conscious awareness [Müştak Sevindik 2025]. These characteristics make *komorebi* well suited to minimizing attentional capture while blending unobtrusively into the environment. It evokes a sense of tranquility and spatial emptiness, known as *yohaku*, accompanied by a gentle perception of time.

Our research suggests that these characteristics are highly compatible with Calm Technology and can be seen as an attempt to embody the design concept of Calm Technology in the form of natural phenomena. This study is not limited to natural light; we treat *komorebi* as a metaphor, including equivalent patterns generated by artificial light sources. We propose a notification method that leverages *komorebi* as a medium within the design framework of Calm Technology.



Figure 1: Natural *komorebi* cast onto an exterior wall (left) and an outdoor table (right); foliage occlusion produces the moving, dappled pattern under sunlight.

2 Prototype Development

We developed prototypes that realize active notification (system-initiated) through a mechanical approach and passive notification (user-initiated) through an optical approach. To this end, we focused on two natural phenomena of *komorebi*. For active notifications, we implemented the "movement" of *komorebi*, which is naturally caused by external forces like wind, using the mechanical approach. For passive notifications, we implemented changes in "shadow hardness", which is caused by the umbra and penumbra resulting from the sun's apparent size, using the optical approach.

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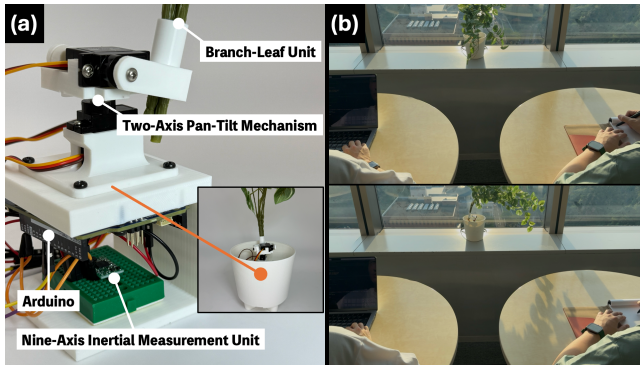


Figure 2: (a) Hardware configuration of the prototype using the mechanical approach. (b) The results of our proposed method. The system mechanically moves the branch-leaf unit to sway and casts komorebi onto the user's hands as a personal notification.

2.1 Mechanical Approach

The mechanical approach utilizes a branch-leaf unit attached to a two-axis pan-tilt mechanism whose orientation can be changed within specified horizontal and vertical ranges (see Figure 2 (a)). This method aims to cast komorebi at a specific location, such as a room with a houseplant near a window. To compensate for changes in solar position over time, the system uses the solar incidence vector to drive the branch-leaf unit and thus control komorebi placement. We compute this vector with a solar simulator (implemented in openFrameworks) from geographic latitude, longitude, and time, combined with device orientation measured by a nine-axis inertial measurement unit. The results are sent via Open Sound Control to an Arduino, which drives the unit. This enables information to be expressed through komorebi patterns of movement and location. For example, it can deliver a personal notification by casting komorebi onto the user's hands (see Figure 2 (b)).

2.2 Optical Approach

The optical approach provides passive notifications, which presuppose a voluntary act of confirmation by the user, through changes in shadow hardness. It involves switching between a bulb-type point light source and a ring-light-type surface light source housed in the same enclosure. The point source creates hard shadows with a distinct umbra and sharp boundaries, while the surface source produces soft shadows with a wide penumbra, resulting in blurred boundaries (see Figure 3 (a)). An Arduino controls the light source switching and brightness via a buck converter. Users can assign meanings, such as room occupancy or the status of a mailbox, to the binary state of hard/soft shadows and check the information at their own convenience, much like looking at a clock (see Figure 3 (b)).

3 Discussion and Future Work

In this paper, we proposed an information notification method mediated by komorebi, based on the design concept of Calm Technology. Through our mechanical and optical approaches, we demonstrated

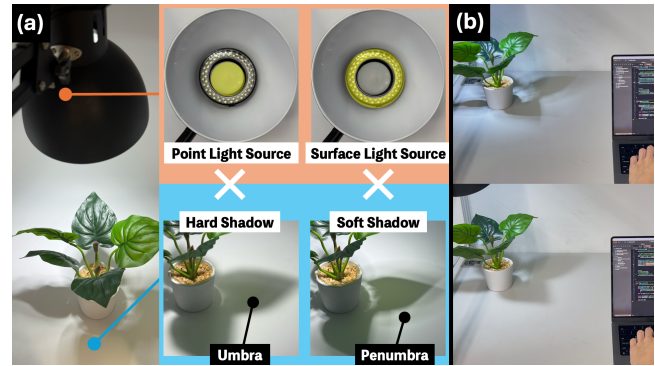


Figure 3: (a) A point light source creates a hard shadow with a distinct umbra, while a surface light source creates a soft shadow with a broad penumbra. (b) In everyday settings, this subtle difference lets users check the system state at their own pace and decide on the next action.

the potential for both active and passive notifications. The novelty of our work, compared to previous research such as Ambient Displays [Wisneski et al. 1998] and Plant Shadow Morphing [Colley et al. 2019], lies in the control of natural optical phenomena like movement and shadow hardness. Furthermore, unlike studies that use plants themselves as a direct medium [Kurihara et al. 2017; Sawaki et al. 2012], our research is distinct in its presentation of information via the casting of light. This study is an initial step toward unlocking new possibilities for information presentation mediated by natural phenomena within the framework of Calm Technology. As future work, we plan to objectively evaluate the effectiveness of our proposed method. We will conduct user studies to verify the extent to which komorebi-based notifications can reduce a user's attentional fragmentation compared to conventional notification methods.

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